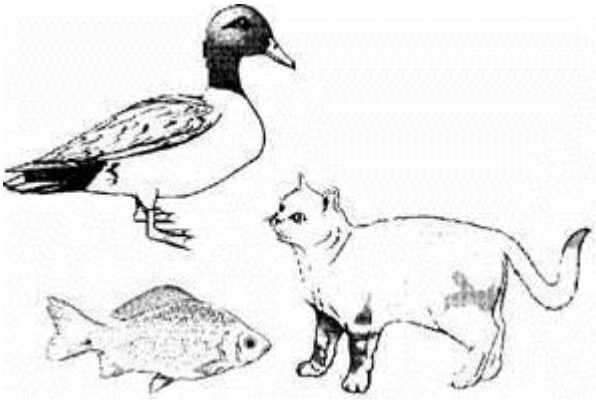
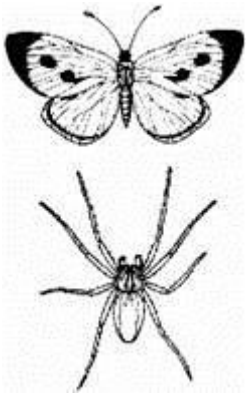
Q1. The list in the box shows four ways to sort animals into two groups.

1.	**They have wings.**	**or**	**They do not have wings.**
2.	**They have a backbone.**	**or**	**They do not have a backbone.**
3.	**They feed their young on milk.**	**or**	**They do not feed their young on milk.**
4.	**They have webbed feet.**	**or**	**They do not have webbed feet.**

Kate, Tom, Dylan and Rania sorted some animals into groups.

(a) Here are Kate's two groups.



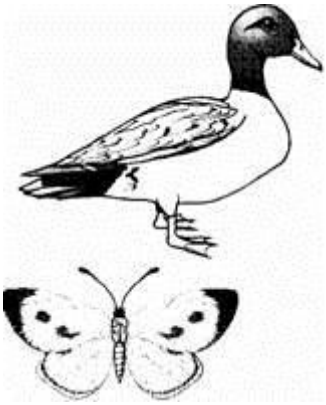


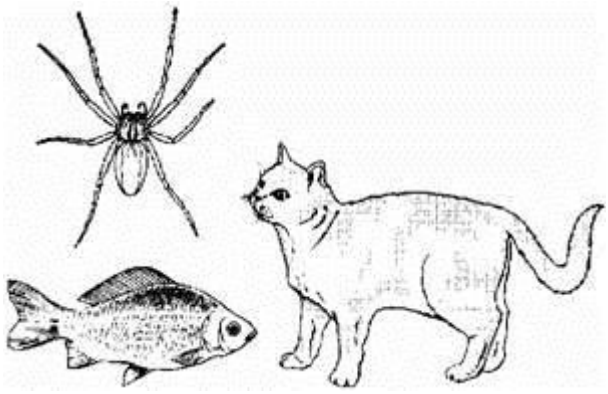
How did Kate sort the animals into groups?

Give the number from the four ways in the box. .....

1 mark

(b) Here are Tom's two groups.



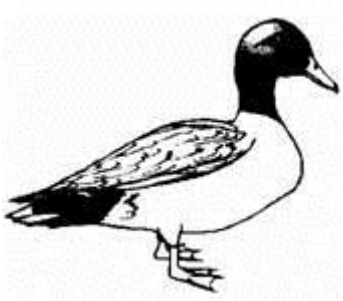


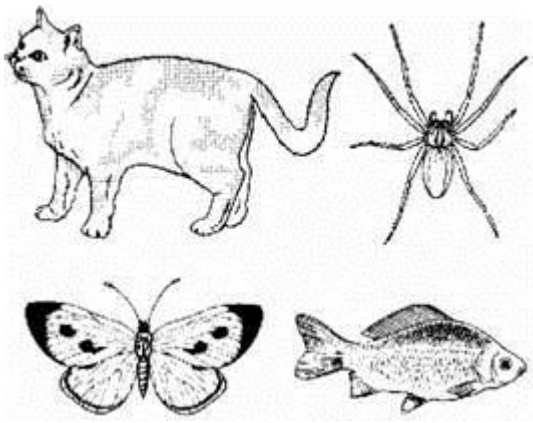
How did Tom sort the animals into groups?

Give the number from the box. .....

1 mark

(c) Here are Dylan's two groups.



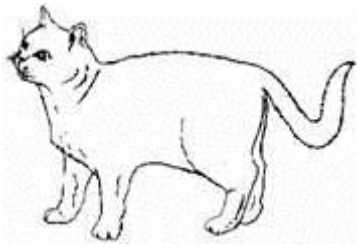


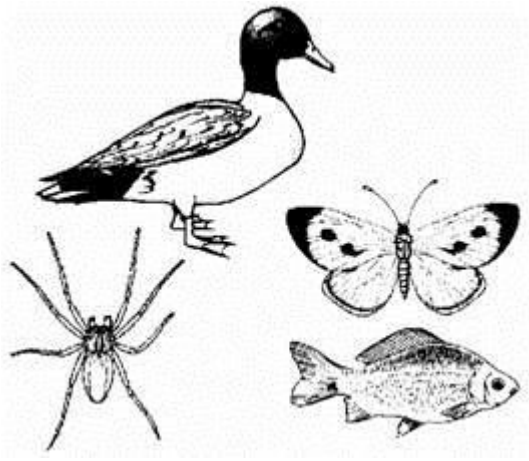
How did Dylan sort the animals into groups?

Give the number from the box. .....

1 mark

(d) Here are Rania's two groups.





How did Rania sort the animals into groups?

Give the number from the box. .....

1 mark

Maximum 4 marks

Q2. The drawing below shows a barn owl.
Barn owls hunt for small animals such as mice.



(a) (i) Look at the drawing of the barn owl.

Give **two** ways the barn owl is suited for catching small animals.

1 .....

2 .....

(ii) Draw a line from each animal below to the word that describes it.
Draw only **two** lines.

animal	word that describes the animal
	predator
mouse	
	prey
barn owl	
	producer

4 marks

(b) The photograph below shows two young barn owls.
They are covered with soft feathers.



Why do the young barn owls need feathers?

.....

.....

(c) Barn owls build nests in farm buildings. Mice eat wheat seeds.

(i) Many old farm buildings have been knocked down so that houses can be built on the farmland.

Give **one** reason why this has caused the number of barn owls to decrease.

.....

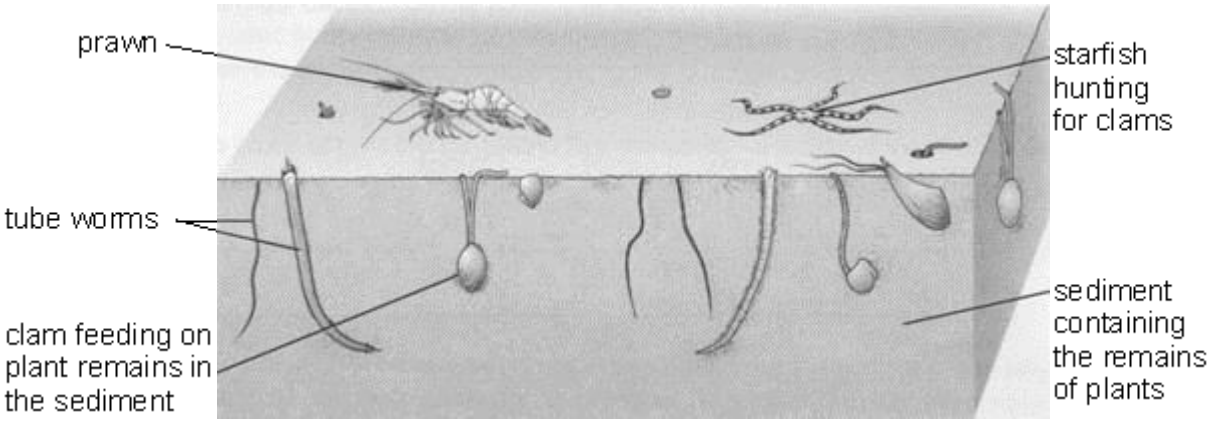
.....

(ii) Suggest **one** reason why farmers like to have barn owls on their farms.

.....

3 marks
maximum 7 marks

Q3. The drawing shows some of the animals which live at the bottom of the North Sea.



not to scale

WWF - UK Data Support For Education Service

(a) Suggest **two** advantages clams get from living in the sediment.

1. ……

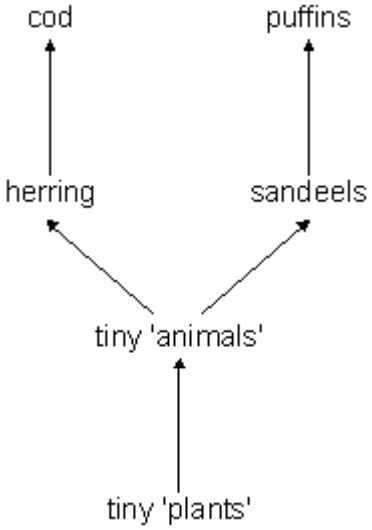
……

2. ……

……

2 marks

(b) Part of a food web in the North Sea is shown below.
Herring, sandeels and cod are types of fish. Puffins are sea birds.



- Herring lay eggs in the gravel on the seabed.

- Sandeels live where the seabed is covered with sand.

Millions of cubic metres of gravel and sand are removed from the bottom of the North Sea every year for roads and buildings.

(i) Give **one** way removing some of the sand and gravel might cause the numbers of herring and cod to decrease.

herring ……

……

……

1 mark

cod ……

……

……

1 mark

(ii) Explain why removing some of the sand has led to a decrease in the number of puffins.

……

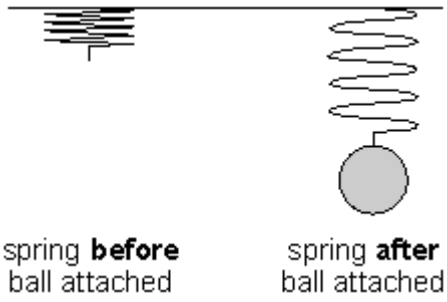
……

……

……

2 marks
Maximum 6 marks

Q4. (a) John attaches a ball to a spring. The diagram below shows what happens.



(i) Which arrow shows the direction of the **force of the ball on the spring**? Tick the correct box.



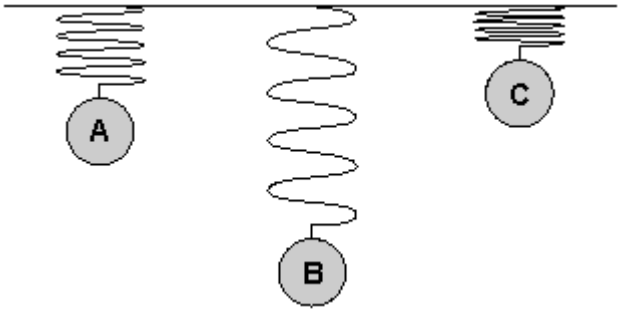
1 mark

(ii) Which arrow shows the direction of the **force of the spring on the ball**? Tick the correct box.



1 mark

(b) The diagram below shows three metal balls attached to **identical** springs.



Which ball is the heaviest?
Write the letter.

.....

1 mark

Explain your answer.

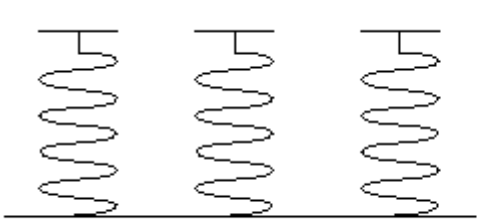
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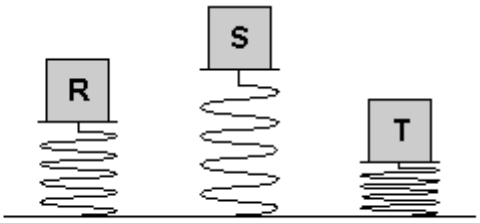
.....

1 mark

(c) John has another three **identical** springs.
He puts a cube on each spring. Each cube has a different mass.

The diagrams below show the springs before and after John added the cubes.


springs **before** adding the cubes



springs **after** adding the cubes

Which cube is the heaviest?
Write the letter.

.....

1 mark

Explain your answer.

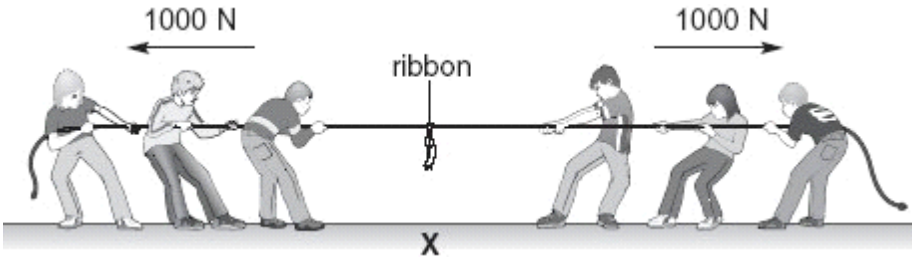
.....

.....

1 mark
maximum 6 marks

Q5. The drawings in parts (a), (b) and (c) show two teams of pupils in a tug-of-war. There is a ribbon tied to the middle of the rope.

(a) The sizes and directions of the forces of each team are shown.



team A **team B**

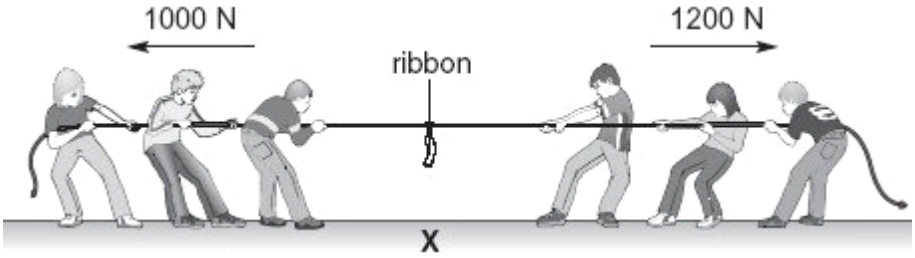
The ribbon stays above point X on the ground.
Give the reason for this.

.....

.....

1 mark

(b) The teams then pull with the forces shown below.

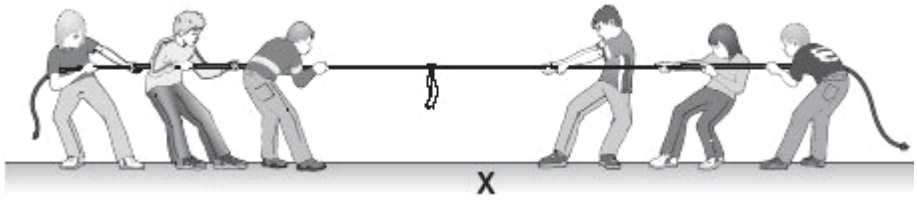


team A **team B**

Draw an arrow on the rope to show the direction in which the ribbon will move.

1 mark

(c) Later, the ribbon was to the left of point X as shown below.



team A **team B**

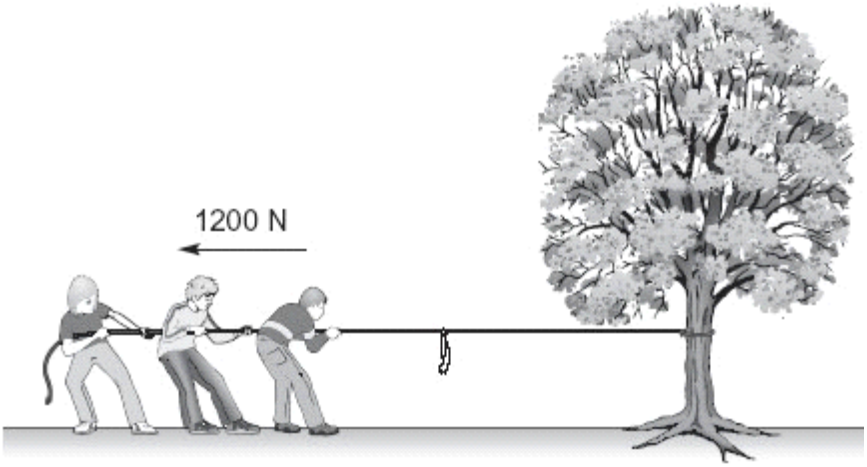
Why did the ribbon move towards the left?

.....

.....

1 mark

(d) Team A practises by pulling a rope tied to a tree.



The team pulls with a force of 1200 N but the tree does **not** move.

What is the force of the tree on the rope?
Tick the correct box.

zero	☐	less than 1200 N	☐	1200 N	☐	more than 1200 N	☐

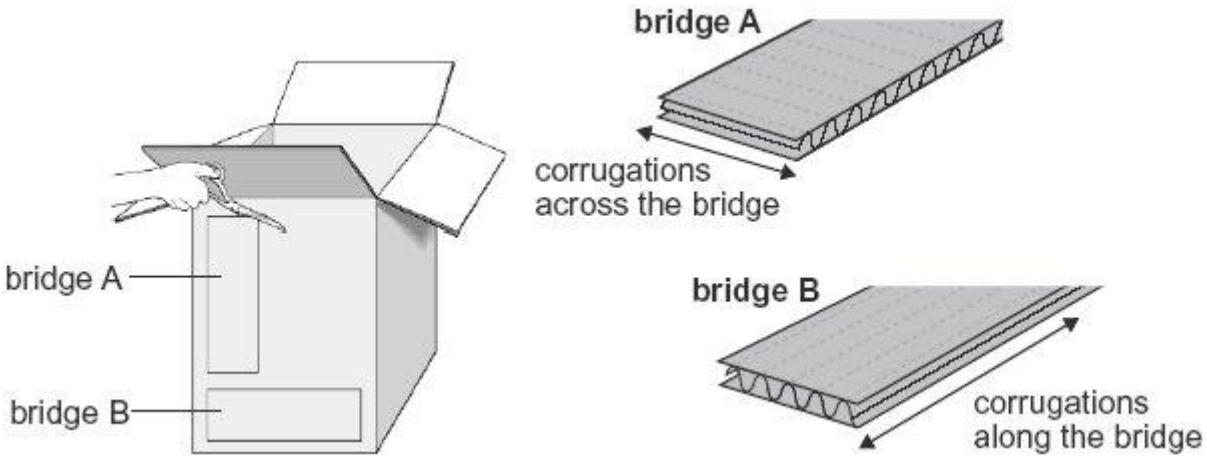
1 mark

(e) The pupils do **not** slip because there is a force between their shoes and the ground. What is the name of this force?

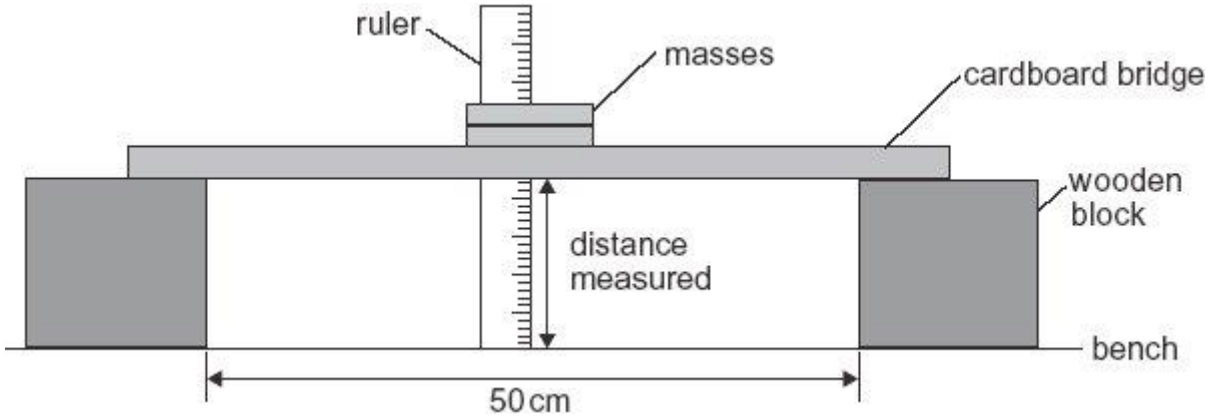
.....

1 mark
maximum 5 marks

Q6. Joe makes two bridges from strips of cardboard cut as shown.



Joe tests the bridges by adding masses to them. He measures the distance from the bench to the bottom of each bridge for different masses as shown.



(a) Suggest **two** things Joe must do to make his test fair.

1. .....

1 mark

2. .....

1 mark

Here are Joe's results.

mass added to bridge (g)	distance from bench to bottom of bridge (cm)	
	bridge A	bridge B
0	7.2	7.2
100	7.1	7.0
200	7.0	6.5
250	6.8	6.1
300	3.0	5.6
350	0.0	5.0

(b) (i) Joe put 325g on each bridge.
Using the results table, estimate the distance from each bridge to the bench.

bridge A ..... cm bridge B ..... cm

1 mark

(ii) Suggest what happened to **bridge A** when it was loaded with 350g.

.....

1 mark

(c) (i) Which bridge would be better for carrying a **200g** toy car?
Tick the correct box.

bridge A ☐ bridge B ☐

Explain your answer.

.....

.....

1 mark

(ii) Which bridge would be better for carrying a **300g** toy car?
Tick the correct box.

bridge A ☐ bridge B ☐

Explain your answer.

.....

.....

1 mark
maximum 6 marks

Q7. Paul had four substances:

citric acid	copper sulphate	indigestion tablet	sugar

He dissolved 1 g of each substance in 20 cm^3 of distilled water.
He used universal indicator to find the pH of each solution.

(a) (i) Sugar solution does **not** change the colour of green universal indicator.

What does this tell you about sugar solution?
Tick the correct box.

It is an acid. ☐ It is an alkali. ☐

It is neutral. ☐ It is sweet. ☐

1 mark

(ii) Suggest the pH of citric acid.

.....

1 mark

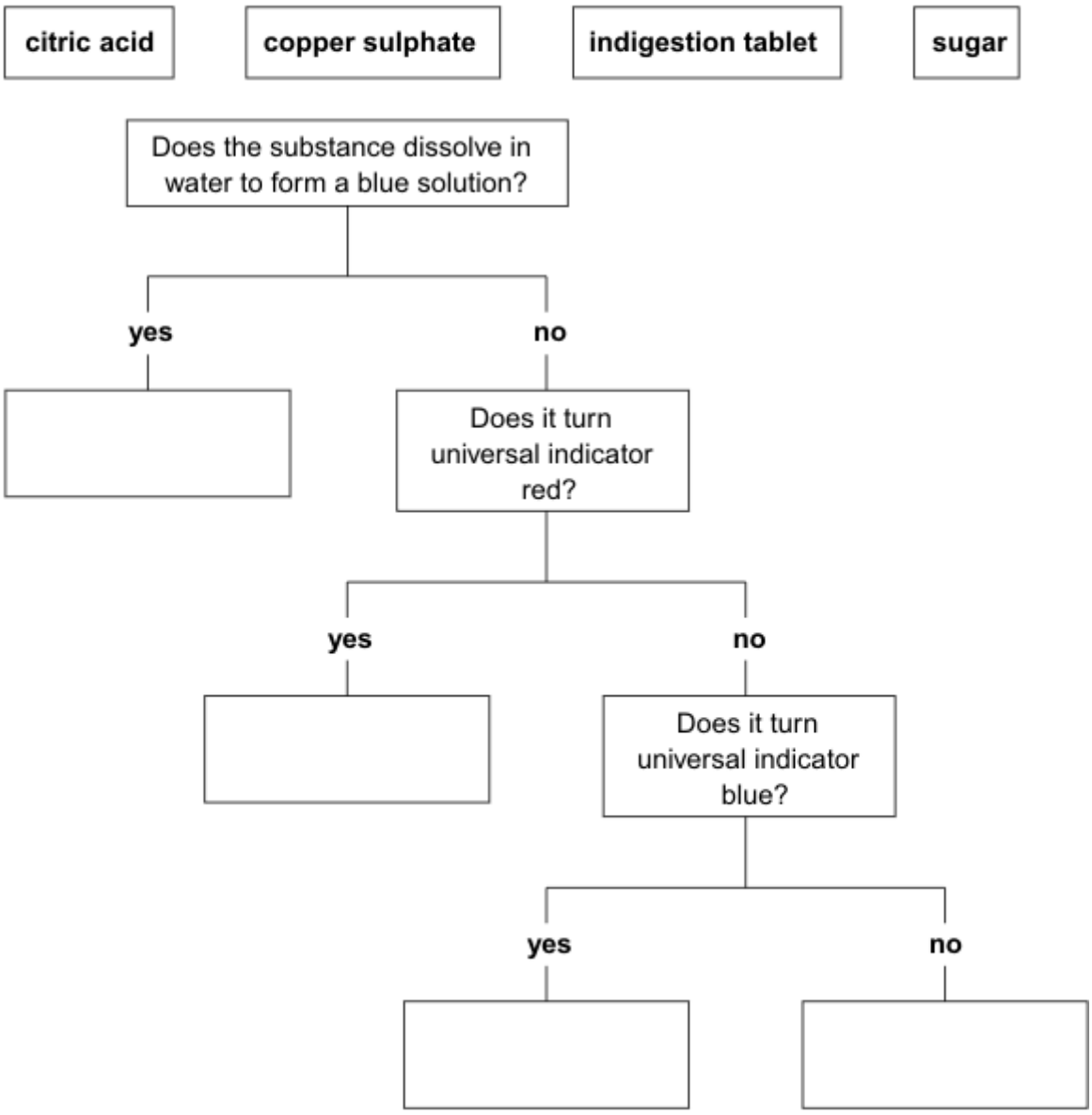
(iii) Indigestion tablets neutralise acid in the stomach.

What does this tell you about indigestion tablets?

.....

1 mark

(b) Complete the flow chart below with the names of the substances in the boxes.



3 marks
maximum 6 marks

Q8. Water from red cabbage can be used to find out if a liquid is acidic, alkaline or neutral.

Type of liquid added to the cabbage water	colour of the cabbage water
acidic	red
alkaline	blue
neutral	purple

John added three different liquids to the cabbage water.

(a) Use the information above to complete the table below.

Liquid added to the cabbage water	colour of the cabbage water	Is the liquid acidic, alkaline or neutral?
water	purple	
lemon juice		acidic
washing up liquid	blue	

3 marks

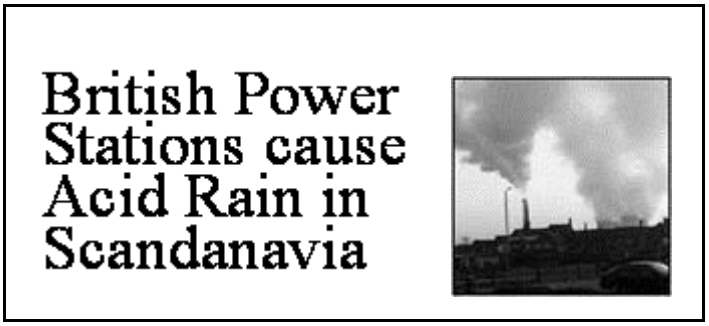
(b) What word describes chemicals which change colour in acids or alkalis?
Tick the correct box.

filters ☐ indicators ☐

liquids ☐ solids ☐

1 mark
Maximum 4 marks

Q9. A headline from a newspaper is shown below.



Some countries claim that acid rain caused by power stations in Britain damages their forests.
Others argue that coal-burning power stations produce cheap electricity and that plants can stand some level of acid rain.

Imagine you are planning a laboratory investigation of the claim:

'plants can stand some level of acid rain'.

Assume you have access to whatever laboratory equipment you need, including:

- seeds
- acid
- seed trays
- soil

Plan a laboratory investigation to test the claim that **'plants can stand some level of acid rain'.**

(a) Name a factor you would need to vary in your investigation.
(This is the independent variable.)

.....

1 mark

(b) (i) What factor would you examine to see the effect?
(This is the dependent variable.)

.....

.....

1 mark

(ii) How could you measure this dependent variable?

.....

1 mark

(c) Suggest **one** factor you would control to ensure that your investigation is fair.

.....

1 mark
Maximum 4 marks

Mark schemes

Q1. (a) 2

1 (L3)

(b) 1

1 (L3)

(c) 4

1 (L3)

(d) 3

1 (L3)

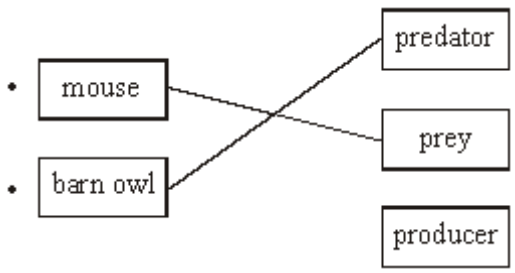
[4]

Q2. (a) (i) any **two** from

- large **or** sharp claws **or** nails **or** talons
  *'claws' **or** 'nails' **or** 'talons' are insufficient*
- large **or** strong **or** sharp beak
  accept 'curved beak'
  'beak' is insufficient
- large eyes or eyes that face forward
  accept 'it can see in the dark'
  'eyes' is insufficient
- wings for hovering
  accept 'it can hover'
  accept 'it can fly silently'

2 (L3)

(ii)



2 (L4)

(b) any **one** from

- to keep warm
  'accept protection from cold'
  *do **not** accept 'to keep the cold out'*
- for insulation
  accept 'for flight'

1 (L3)

(c) (i) any **one** from

- fewer places to nest
  *accept 'nowhere to nest' **or** 'no home'*
- fewer places to lay eggs
  accept 'nowhere to lay eggs'
- they lay fewer eggs
  accept 'they cannot lay eggs'
- less food for owls
  *accept 'fewer mice' **or** 'no mice' **or** 'no food'*
- owls can be frightened by traffic **or** people
  accept 'it is too noisy'
- loss of habitat

1 (L4)

(ii) they catch **or** kill mice
accept 'to stop mice eating the crops'
accept 'to keep mice away'
*'to protect the crop **or** grain' is insufficient*

1 (L4)

[7]

Q3. (a) any **two** from

- they live in **or** close to their food
  accept 'food is available'
- they are not washed away
- protection
  accept 'they cannot be seen'

2 (L6)

(b) (i) any **one** from

- there are fewer eggs
  accept 'herring lay fewer eggs'

*do **not** accept 'they cannot lay eggs'*

- fewer herring eggs hatch
- herring eggs are removed
  accept 'they have fewer places to lay their eggs'

1 (L5)

(ii) • cod have fewer herring to eat **or** less food
*do **not** accept 'they have no herring to eat'*

1 (L5)

(iii) • the habitat of sandeels is reduced **or** destroyed **or** removed
accept 'sandeels have nowhere to live'

1 (L5)

- puffins have fewer sandeels to eat
  accept 'there are fewer sandeels'
  accept 'sandeels are removed'

1 (L5)

[6]

Q4. (a) (i) • ↓ ✓

1 (L3)

if more than one box is ticked, award no mark

(ii) • ↑ ✓
if more than one box is ticked, award no mark

1 (L3)

(b) • B
*accept 'the middle **or** second one'*

1 (L3)

any **one** from

- it hangs down the most
  accept 'the spring is wider apart'
- it stretches the spring most
  accept 'it stretches more'
- the spring is longer
  accept 'it stretched the longest'
  this mark cannot be awarded if the response conflicts with the first part of (b)
  a comparative answer is required for the mark
  non-comparative answers such as 'it is long' are insufficient

1 (L3)

(c) • T

accept 'the last one'

1 (L3)

any **one** from

- the spring was pushed down the most amount

  accept 'it squashes more'

- the cube pushed it down the most

  accept 'the spring is the tightest'

- the spring is shorter

  accept 'it is shortest'

  this mark cannot be awarded if the response conflicts with the first part of (c)
  a comparative answer is required for the mark
  non-comparative answers such as 'it is short' are insufficient

1 (L4)

[6]

Q5. (a) any **one** from

- the forces are balanced

  ignore references to gravity if the answer is in terms of balanced forces

- the forces are equal **or** the same

  'the sides are equal' is insufficient

- the forces are both 1000 N

  accept 'the forces are both 1000'
  accept 'the newtons are even'
  *do **not** accept 'both teams weigh 1000 N'*

- they pull with the same force **or** equally hard

  accept 'both teams have the same strength'

1 (L4)

(b) an arrow drawn to the right

accept an arrow drawn to the right anywhere on the drawing

1 (L3)

(c) any **one** from

- team A pulled harder than team B

  *accept 'team A pulled harder' **or** 'team A pulled more'*
  ***or** 'they pulled harder'*
  accept the converse

- team A was stronger

  accept 'they used more strength'

- team A was pulling with more than 1000
- team B was pulling with less than 1000
- there was more force to the left

accept 'there are more newtons to the left'

1 (L4)

(d) 1200 N ✓

if more than one box is ticked, award no mark

1 (L4)

(e) friction

1 (L4)

[5]

Q6. (a) any **two** from

- same type of cardboard

accept 'same cardboard or box'

- same width (of cardboard)
- same length (of cardboard)
- same thickness of cardboard pieces

accept, for one mark, 'the same sized bridge'
if not given with 'width' or 'length' or thickness
accept 'equal-sized cardboard pieces'

- loaded in the centre
- same (height of) blocks

accept 'same height at the start'
'same height of bridge' is insufficient

- same distance between blocks
- measure in the same place

accept 'leave masses on for the same amount of time'
accept 'masses of the same shape or type'

2 (L5)

(b) (i)
- *bridge A:* any number from 0.0 to 2.9
  bridge B: any number from 5.1 to 5.5

***both** answers are required for the mark*

1 (L5)

(ii) any **one** from

- it collapsed
- it broke
- it folded
- it reached the bench

1 (L5)

(c) (i) • bridge A ✓

if more than one box is ticked, award no mark
***both** the bridge and a reason are required for the mark*

any **one** from

- it bends less (at 200 g)

  accept 'there is a bigger gap to the bench'
  'it is stronger' is insufficient

- bridge B bends more

  accept 'it is higher'
  accept 'bridge A is 7.0 cm and bridge B is only 6.5 cm'
  'bridge A is 7.0 cm and bridge B is 6.5 cm' is insufficient

1 (L6)

(ii) • bridge B ✓

if more than one box is ticked, award no mark
***both** the bridge and a reason are required for the mark*

any **one** from

- it bends less (at 300 g)

  accept 'there is a bigger gap to the bench'
  'it is stronger' is insufficient
  accept 'it is higher'
  *accept 'bridge B is 5.6 cm and bridge A is **only** 3.0 cm'*
  'bridge A was 3.0 cm, bridge B was 5.6 cm' is insufficient

- bridge A bends more

  accept 'bridge A is about to break'
  'it is about to break' is insufficient

1 (L6)

[6]

Q7. (a) (i) • It is neutral ✓

if more than one box is ticked, award no mark

1 (L5)

(ii) • any number less than 7

1 (L6)

(iii) any **one** from

• they are alkaline **or** an alkali

accept 'a soluble base'
accept 'it has a pH above 7'
specific pH values such as 'its pH is 8.5' are insufficient

• they react with the acid

1 (L5)

(b) • copper sulphate

accept 'copper sulfate'
accept 'the sulphate' for 'copper sulphate'

• citric acid

accept 'acid' for 'citric acid'

• indigestion tablet • sugar

accept 'tablet' for 'indigestion tablet'
*if **all four** answers are correct, award three marks*
*if **two or three** answers are correct, award two marks*
*if **one** answer is correct, award one mark*

3 (L5)

[6]

-

Q8. (a)

Liquid added to the cabbage water	Colour of the cabbage water	Is the liquid acidic, alkaline or neutral?
water	purple	neutral
lemon juice	red	acidic
washing up liquid	blue	alkaline

3 (L4)

(b) indicators

if more than one box is ticked, award no mark

1 (L3)

[4]

Q9. (a) any **one** from

- plants subjected to **or** not subjected to acid
- pH of the acid
- strength of solution

  accept 'concentration of acid'
- volume of acid

  accept 'amount of acid'

  accept any appropriate independent variable

1 (L7)

(b) (i) any **one** from

mark parts (b) (i) and (b) (ii) together

- plants live **or** die
- plants healthy **or** not healthy
- plants **or** leaves change colour
- how many seeds grow

1 (L7)

(ii) any **one** from

the dependent variable must relate to the independent variable mentioned in part (a)

- number of plants dying **or** ailing

  accept 'count them'

  accept a reference to appropriate measuring equipment
- number of leaves falling **or** ailing
- mass of plant matter
- area of plant leaf growth
- height of plant

1 (L7)

(c) any **one** from

- soil nutrients
- temperature
- humidity
- light
- acidity of soil at the beginning

accept any suitable control relevant to the factors specified in parts (a) and (b) (i)

1 (L7)

[4]